

Exploring the Use of Large Language Model-Driven Chatbots in Virtual Reality to Train Autistic Individuals in Job Communication Skills

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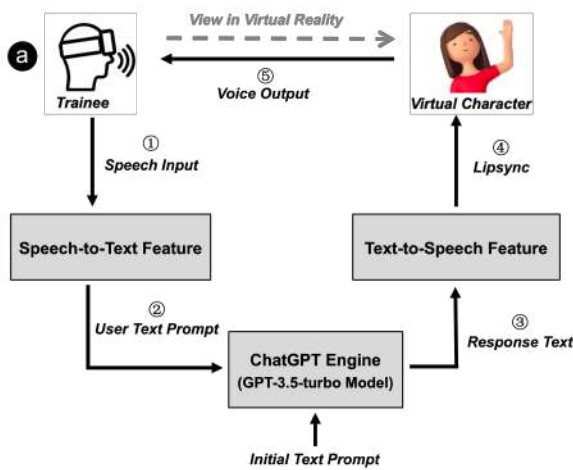


Figure 1: An overview of the VR chatbot application's setup. (a) The setup of the VR chatbot application for communication skills training. (b) A screenshot of a scenario where the virtual character is interacting with a trainee.

ABSTRACT

Autistic individuals commonly encounter challenges in communicating with others which can lead to difficulties in obtaining and maintaining jobs. Thus, job training programs have emphasized training the communication skills of autistic individuals to improve their employability. Hence, we developed a virtual reality application that features avatars as chatbots powered by Large Language Models (LLMs), such as GPT-3.5 Turbo, and employs speech-based interactions with users. The use of LLM-driven chatbots allows job coaches to create training scenarios for trainees using text prompts. We conducted a preliminary study with three autistic trainees and two job coaches to gather early-stage feedback on the application's usability and user experience. In the study, the trainee

participants were asked to interact with the application in two scenarios involving customer interactions. Our findings indicate that our application shows promise for training job communication. Furthermore, we discuss its user experience aspects from the trainees' and job coaches' perspectives.

CCS CONCEPTS

• **Human-centered computing** → Empirical studies in accessibility; User studies.

KEYWORDS

Accessibility, Autism Spectrum Disorder, Virtual Reality, Job-related Soft Skill, Communication Skill Training, Generative AI

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1 INTRODUCTION

Autism is a lifelong neurodevelopmental condition that significantly impacts individuals in various aspects of life [36]. Autistic individuals commonly encounter challenges in social interactions, communication, and learning, which can manifest in distinct characteristics and challenges [25]. These unique traits and challenges associated with autism have been found to contribute to a range of educational and employment obstacles faced by individuals on the spectrum [14, 24, 28, 29].

To improve the employability of autistic people, various educational and vocational rehabilitation programs have been developed focusing on job-training autistic people with the help of job coaches [13]. While many of these programs focus on training the hands-on skills of autistic trainees, job coaches in these programs (people who train them for employment) also emphasize the importance of training the communication skills of their autistic trainees such as communicating with customers, confidently facing interviews, etc. [2, 14, 29]. To train the communication skills of an autistic trainee, job coaches employ a variety of methods primarily involving simulated one-on-one practice “chat” sessions for different scenarios. However, developing new social communication scenarios, continuously conducting practice sessions, etc., can be taxing for a job coach and a trainee will always require a partner to practice such scenarios.

Recent research has begun exploring novel approaches that integrated Artificial Intelligence (AI)-powered chatbots with Virtual Reality (VR) [1, 3, 5, 23, 31] to offer more personalized training opportunities in job training. In these works, the AI-powered chatbots simulate communication experiences in various settings including virtual role-play for job interview simulations [30, 32], training assistance [33, 35], and emotional support [4, 8, 21, 22]. Here, while VR can provide an immersive and customizable environment for trainees to use and practice independently [5], training AI models and adapting the models for custom scenarios has been a challenging task [39]. However, the recent introduction of pre-trained Large Language Models (LLMs) has enabled more dynamic chatbot applications, such as ChatGPT [26], that simulate natural conversations based on text based instructions called ‘prompts’. Thus, to address the above challenge, we developed a chatbot application driven by an LLM (the GPT-3.5 Turbo model) to assist autistic individuals in practicing communication skills in VR (Figure 1(b)).

Additionally, while many studies present the perspectives of the autistic trainees, limited studies have assessed the views of the job coaches who work with the autistic individuals to understand their perspectives on using LLM-driven chatbots in VR for job communication skills training. Inspired by the previous works, our approach aims to fill this gap by focusing on gathering feedback from both autistic trainees and job coaches across different virtual service provider-customer interaction scenarios. To gather early-stage feedback on our approach and how it can assist job coaches in training autistic individuals of their job-related communication skills, we conducted a preliminary study involving three autistic trainees and two job coaches. In this study, the trainee participants were asked to interact with avatars in VR environments to explore this technique’s socializing and communication experience in two job scenarios: taking orders in a cafe and handling complaints about

item returns in a shop. Meanwhile, the job coach participants were asked to evaluate the trainees’ experience and discuss how this technique could be further improved and integrated into the job training practices. Here, we posed two research questions: RQ1: “*What is the user experience of the VR chatbot application in job communication skill training?*” and RQ2: “*What are the job coaches’ perspectives on using the VR chatbot application for job communication skills training?*”

Our contributions of this work are:

- 1) We developed a VR chatbot application powered by the GPT-3.5 Turbo model that allows job coaches to create job training scenarios that can simulate an immersive service provider-customer communication experience.
- 2) We conducted a preliminary study to assess the user experience of our application in two job communication training scenarios from the perspectives of three autistic trainees and two job coaches who work with people with disabilities.

2 VR CHATBOT APPLICATION

To facilitate training communication skills of autistic trainees, we developed a VR chatbot application powered by GPT-3.5 Turbo. The main goals of this application were to allow the job coaches to flexibly and efficiently create scenarios and for trainees to immersively train their communication skills.

2.1 Application

The VR chatbot application was developed using the Unity Platform version 2022.3.7f1 and Oculus Integration Package version 54.1. This application runs on a Windows laptop and is viewed on a Meta VR headset through Meta Quest Link.

A scene in the application consists of a virtual character and a virtual environment (Figure 1(a)). The virtual character, developed using the Meta Avatars SDK¹, which supports using a mock-up output to generate life-like facial expressions and eye movements, aims to deliver an engaging experience to autistic individuals [6, 10, 20]. Additionally, it uses Oculus Lipsync for Unity² to generate virtual characters’ lip movements based on the input speech sound. The virtual environment is presented using a 360-degree photo captured from the real world to simplify the scenario setup for job coaches while providing users with an immersive experience.

To offer autistic individuals a realistic communication experience with the virtual characters [20], the application only accepts voice-based interactions as the input, as shown in Figure 1(a). Users initiate this speech interaction by pressing a button on the VR headset controller. Once the user’s speech is recorded, the application transcribes the voice input using Meta Voice SDK’s Dictation feature³. Next, the transcript is sent as text strings to the GPT model through OpenAI APIs⁴. We used the GPT model as it is a natural language generation engine that can also simulate conversations [34]. Meanwhile, it allows implementing different scenarios easily through initial text prompts (system prompts) [12, 37] where the virtual character’s dialog characteristics can be influenced based

¹<https://developer.oculus.com/documentation/unity/meta-avatars-overview/>

²<https://developer.oculus.com/documentation/unity/audio-ovrlipsync-unity/>

³<https://developer.oculus.com/documentation/unity/voice-sdk-dictation/>

⁴<https://openai.com/blog/openai-api>

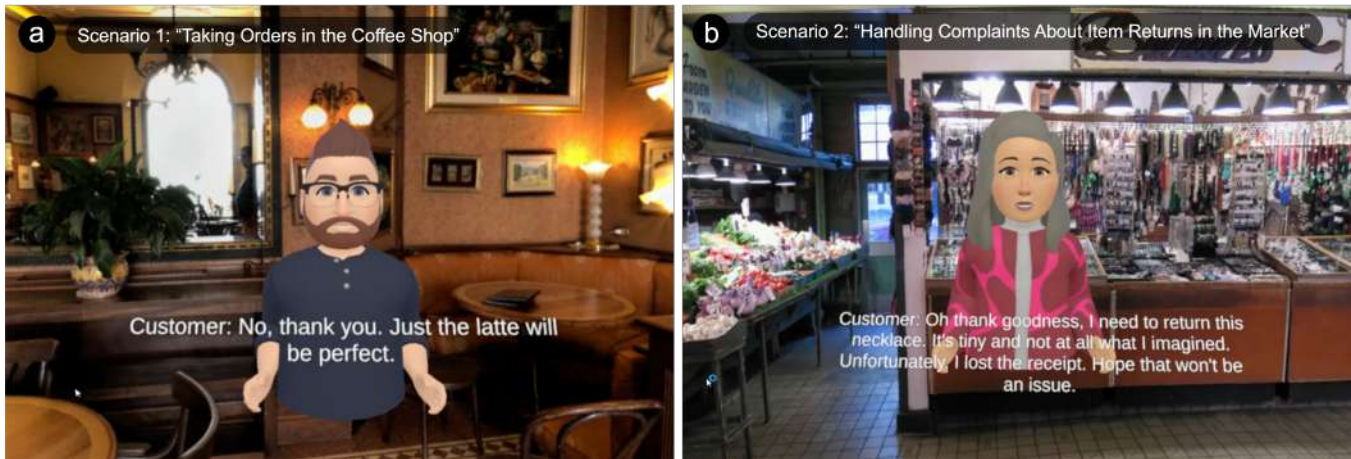


Figure 2: The in-app views of the two scenarios within the VR chatbot application. (a) Scenario 1: Users act as coffee shop owners and take orders from customers. (b) Scenario 2: Users act as jewelry store owners and handle customer complaints regarding item returns.

on the instructions of the prompt text. In the current application version, the initial prompt that sets the scenario is stored in an editable text file within the application folder (see Sec 2.2). We used the “gpt-3.5-turbo-16k-0613” model developed by OpenAI for our study. This version of GPT model was selected based on its availability at the time the study was conducted, its immediate response time, excellent natural language understanding and generation capabilities, and high maximum token capacity [18]. To assist the model in understanding and remembering the dialog context, previous conversation dialogues are sent to the GPT model along with the transcript of the user’s latest speech input.

Next, the response message obtained from the GPT model is converted to speech using Meta Voice SDK’s Text-to-Speech (TTS) feature, which is powered by a Wit.ai-based service⁵. Especially during the coaching process, the dialogues from either the user or the virtual character are displayed as captions on the screen. This allows job coaches who observe the application’s view from an external screen to understand the conversations more clearly, thereby enhancing the coaching experience. For records and analysis, the conversations are saved on a local text file on the PC. Due to the 20-second maximum audio length limit of Wit.ai’s TTS feature, we determined through trial and error that a safe maximum text string length for the virtual character would be approximately 180 characters. This limitation is included as part of the initial prompt as a response restriction.

2.2 Scenarios

For this preliminary study (Sec 3), we designed two scenarios to simulate communication experiences in real-world service provider-customer interaction scenarios. These scenarios were proposed and designed during prior discussions with job coaches from a local job training organization.

In *Scenario 1* (see Figure 2(a)), the user assumes the role of a coffee shop owner and takes orders from a virtual character posing

as a customer. The expected ending criteria for this scenario are that the participant can ask for the ordered items, accept money, and hand over the prepared order to the customer.

In *Scenario 2* (see Figure 2(b)), the user takes on the role of a jewelry store owner and handles a complaint from a virtual customer who wishes to return a necklace without a receipt. In this scenario, we expected the participant to listen to the customer’s request, comfort the customer, or address the customer’s concern.

To set up these two scenarios, we have built two distinct scenes in our VR chatbot application: a coffee shop and a market. In both scenes, a virtual character stands at the center, facing the user, and is surrounded by a 360-degree photo that serves as the environment. What differentiates the two scenes are the initial prompts, the virtual character’s appearance and voice, as well as the background image. The initial prompts for both scenarios were refined through trial and error.

Scenario 1 used a virtual character with a preset speech voice named “Prospector” from the Voice SDK’s TTS feature. Its backdrop was a 360-degree image of a coffee shop interior. We defined its initial prompt as “Act as a cafe customer. You are ordering something from a coffee shop. Make your sentence as oral as possible. Do not exceed 180 characters. Now let’s start acting.”

Scenario 2 used a virtual character with a preset character speech voice named “Rebecca” from the TTS feature. Its backdrop was a 360-degree image of a marketplace with small stalls. We set its initial prompt as “Act as a complaining customer. You are returning a necklace without a receipt. Make your sentence as oral as possible. Do not exceed 180 characters. Now let’s start acting.”

3 METHODOLOGY

To explore and evaluate the user experience of our approach, we conducted a preliminary study that involved autistic trainees and job coaches.

⁵<https://developer.oculus.com/documentation/unity/voice-sdk-tts-overview/>

3.1 Participants

We recruited three job trainees (TP1-TP3) aged 26 to 63 (Mean = 38.67, SD = 21.08. 3 males) and two job coaches (JCP1, JCP2) with a mean age of 47 (SD = 8.49. 1 male and 1 female), as the participants of this study through word of mouth from a local job training organization that works with people with disabilities. One job coach worked with two trainees for six months, while another worked with the third trainee for the same period, providing training for their employment. All three trainee participants were diagnosed with autism and they all can communicate, read, comprehend interview questions, and perform the required training tasks using our proposed VR application. Among the trainee participants, TP3 was the only one who reported previous experience with VR, as well as having prior job and job training experience. Regarding the two job coach participants, JCP1 has been working as a job coach for 15 years and has coached 15 individuals with intellectual disabilities. JCP2 has been working as a job coach for 2 years and has experience with 8 individuals with intellectual disabilities. Each participant was compensated with 30 USD for their participation.

3.2 Apparatus

The study was conducted in a meeting room at the job training organization. To minimize the risk of VR-related motion sickness, participants were advised to remain seated during the VR experience. The equipment used included a Meta Quest headset, a 15-inch Windows 11 laptop, and a 55-inch television. The previously described scenarios were employed throughout the study. During the sessions, the VR headset was connected to the laptop, which mirrored the headset's view. This view was also projected onto the television, allowing job coaches to closely monitor and guide the trainee's experience. The laptop's screen was video-recorded and participants' conversations were voice-recorded. Additionally, transcripts of conversations between participants and chatbots were documented and stored on the laptop.

3.3 Study Procedure

The study was approved by the ethics review board of the authors' institution. All five participants (trainees and job coaches) participated in the same initial study session (pre-study questionnaires and VR experiences), followed by a focus group session with the trainees and an interview with the job coaches. This format was adopted to allow job coaches to assist and instruct the trainees as required.

The trainees' study session contained a pre-study survey, a VR experience study, and a focus group. After giving an introduction and having the participant sign the informed consent form, we asked the participants to complete the pre-study questionnaire, which inquired about trainees' demographic information, professional background, and previous user experience with VR. Next, we had the trainees sequentially experience two virtual customer interaction scenarios in VR, starting with Scenario 1 followed by Scenario 2. This order was chosen based on the complexity, starting with the simpler scenario featuring fixed processes, and progressing to the more complex one without specific handling procedures. When each trainee participants was experiencing a scenario, both job coaches also stayed in the meeting room to observe and guide

the trainee. Here, the trainees were asked to "communicate" with the customers similar to a typical practice session during their communication training. Thus, the job coaches were able to stand near their trainees, watch trainees' views within the application as projected on the television, and provide verbal instructions to them. There was a 5-minute break after experiencing each scenario. Next, a focus group study was conducted among all the trainees. Here, we aimed to collect feedback on three key aspects: the qualitative user experience with the VR chatbot application, suggestions for improving the VR chatbot application, and their expectations for using the VR chatbot application in job communication skills training. Overall, the trainees' study session took no more than 60 minutes to complete.

The study session for job coaches included four components: a pre-study survey, observation during the trainee's VR experience study, observation during the trainee's focus group study, and a post-study interview. Similar to the trainee's session, after giving an introduction and having the job coach participants sign the informed consent form, they were asked to complete a survey to collect their demographic information and their expectations about communication skills training via chatbots in VR. Subsequently, as stated above, job coach participants joined the experiment session to observe and guide the trainee participants. Then, they participated in a focus group, where they listened to the trainee participants' responses. After all the trainees finished their study sessions, we conducted a semi-structured post-test interview with the job coaches. Before the interview started, the job coaches had the option to personally experience the two virtual customer interaction scenarios in VR, to better understand the simulation experience. During the interview session, the job coaches were asked to provide feedback on several areas: the job training experience of the trainees with intellectual disabilities, suggested improvements for the current application, and their visions for using this approach in future communication skills training programs. The interview took less than 30 minutes to complete.

4 FINDINGS

We conducted a thematic analysis [11] of the conversation transcripts between the chatbot application and the participants, the observations recorded during the study, as well as of the feedback collected from the participants. By analyzing these data, we identified several themes reflecting the participants' experiences with this approach.

4.1 Role-Playing Simulations Using the VR Chatbot Application

According to our analysis of the transcripts, our application that leverages LLM-driven chatbots demonstrated abilities in understanding the context, proposing creative conversations based on the scenario, and keeping the conversation going. These abilities benefit the performance of the job training-oriented role-playing simulations of our study. For example, in Scenario 1, the chatbot application can propose a specific order request, *"Can I get a medium latte to go with almond milk, please? Oh, and a blueberry muffin too, if you have any left!"* This was in response to our initial prompt that

instructed without being given much context (See Sec 2.2). In Scenario 2, the chatbot application showed it could adapt the scenarios in response to user inputs. When TP2 tried to verify if the item was brought from their store, the chatbot application could adapt to the context to keep the conversation going: *“Yes, I’m positive! I remember the salesperson recommending it to me. But now it’s broken and I want my money back, even without a receipt!”*

Despite its creativity, the chatbots can generate emotionally charged responses, aiming to create conversations that more closely resemble real-world interactions. This was particularly evident in Scenario 2. When TP1 mentioned their “manager” was too busy to offer assistance, the virtual avatar responded in an agitated manner: *“Of course, your manager is ‘busy’. This is just adding to the terrible service I’ve received. I will be taking my business elsewhere from now on.”*

4.2 User Experience and Effectiveness of Using the Application for Communication Skills Training

In our study, the trainee participants found their overall experience with the VR chatbot application engaging and comfortable. They reported not feeling severe motion sickness during and after their experience with the application, which might be attributed to the fact that participants interacted with the chatbots while seated. Additionally, they were impressed by the level of immersion in the scenario environments. *“I felt like being in the store,”* said TP1.

During the focus group, participants gave positive feedback regarding the clarity of the conversations with the virtual character and how the training process with the application could promote thinking and enhance communication skills. Here, the trainee participants noted the ability to pause between conversation turns allows them to think carefully before responding to “customers”, offering a unique training experience compared to real-world communication scenarios. As TP3 put it, *“I was trying to handle the situation, in terms of what I was going to talk to her (the chatbot),”* explaining his pause before replying to the customer.

Feedback from job coaches affirmed the effectiveness of the VR chatbot application. Both job coaches were enthusiastic about incorporating this technology into their training sessions. Especially, JCP2 noted the inherent challenges of communication skills training and the promise of the VR chatbot application in the skills training, stating that *“As every situation is very different, so it is difficult to handle these situations like diverting to a manager or what else can be done. In the real world, you have to be on your toes, so the dynamic nature of this tool is going to be very helpful.”*

Additionally, JCP1 noted that this application enabled trainees to independently practice their communication skills. He emphasized that the VR environment assists the autistic trainees in concentrating more effectively, by reducing distractions from their surroundings. Both job coaches agreed that generating transcripts or recording sessions during the training process were useful, as it allowed for a more effective evaluation of the trainees’ progress and outcomes.

4.3 Expectations of Job Coaches and Proposed Improvements

In discussing the application’s setup, JCP1 emphasized the need for simplification to accommodate job coaches who may not be tech-savvy. JCP1 suggested that implementing a more user-friendly scenario setup, perhaps through a PC-based integrated app, would be beneficial. This would enable coaches to develop and evaluate training scenarios without the need to interact directly with a VR device. JCP1 stated, *“I hope that we, as job coaches, could develop a training scenario easily using this tool, even for those who are unfamiliar with VR and specific text-based instructions (like GPT prompts).”*

Meanwhile, the job coaches highlighted the training process with the application could be optimized. For instance, in Scenario 2, it was observed that when TP2 and TP3 were undergoing the scenario, the “customers” consistently complained about their shopping experience. This led to their conversations deviating from the intended direction until the job coaches stepped in to guide the trainees. Thus, we propose incorporating specific ending criteria when designing scenarios using initial text prompts. For example, in Scenario 2, an appropriate ending criterion could be the “customers” successfully returning their products.

5 DISCUSSION

To answer RQ1, as indicated by the trainee participants, the VR chatbot application creates a sense of presence in the given job environment. It also simulates a realistic and engaging service provider-customer communication experience in the scenarios. These comments might be related to the adoption of LLM-driven chatbots and VR in the application. Additionally, job coach participants noted that VR helps autistic trainees focus better on the training tasks, as the autistic individuals are often easily distracted by their surroundings in typical training process [7].

Meanwhile, the trainee participants also agreed on the application’s effectiveness in practicing communication skills. They found the ability to pause and think about their responses to the virtual character, who acted as a “customer,” to be beneficial for their communication skills training as it can assist in promoting them thinking.

To answer RQ2, according to feedback from job coaches, there are inherent difficulties in traditional communication skills training. One of the significant challenges is coaching trainees to handle the various situations that arise in a job scenario. In this context, the adoption of LLM-driven chatbots in the application could be particularly beneficial. It can dynamically generate virtual customer responses and reactions based on a trainee’s input and the given scenario design prompt, which can simplify the communication training setup and process for job coaches.

Moreover, job coaches expressed expectations to enhance the coaching experience with our application. In addition to the desire to simplify the setup process for VR training, which was also noted in previous studies in similar and other contexts [9, 17, 27], our observation and job coaches’ feedback highlighted the importance of specifying prompts in scenarios design. In our practice in this preliminary study, two simple prompts are used for setting up the two job scenarios. This approach aimed to provide more exploration

space for using the application in job communication training and to gather insights from autistic trainees and job coaches. However, due to the natural language generation characteristics of the LLM-driven chatbots, a prompt without specified context might be more likely to generate unrealistic responses based on the trainees' input and fail to fulfill the intended purpose of a training scenario [16, 19]. This issue was particularly evident during Scenario 2, which had ambiguous requirements in its prompt design. Therefore, when using LLM-driven chatbots in job communication training, especially with autistic individuals, it is necessary for job coaches to use more explicit prompts [37]. These should include clear requirements and specific ending criteria to achieve better training outcomes.

Additionally, although this study was conducted in a controlled environment, where job coaches were present to observe and guide the trainee participants, we did not have specific restrictions on the agents' behaviors. This was mainly because prior to our study, the job coaches expressed their interests in exploring everyday service provider-customer interactions through this simulation. However, the job coaches highlighted that, when these chatbots are used in more independent training sessions, they may need to guide trainees in managing those unrealistic responses (which often referred to as "hallucination") from the LLMs [16]. Meanwhile, it is also crucial for them to protect trainees from potential ethical issues and ensure their overall well-being during the training process [15, 38].

6 LIMITATIONS AND FUTURE WORK

This preliminary study included only three trainee participants and two job coaches, as several other participants withdrew due to personal reasons. Furthermore, recruiting participants of this particular demographic (autistic trainees and job coaches who work with them) has been challenging due to similar personal constraints. To address this, we plan to collaborate with more local job training organizations, thereby enlarging our participant pool.

Meanwhile, this study was conducted in a controlled environment over a two-hour session. To further explore the effects of our approach, we plan to conduct longitudinal studies. These future studies will primarily involve job coaches creating scenarios tailored to their needs using our system and integrating this technique into regular communication skill training sessions. These future studies will enable us to gather more comprehensive feedback.

In terms of experience with avatars, currently, the avatars can converse robustly with participants, but their physical expressions are limited to lip movements and blinking. Therefore, our future research will aim to investigate the integration of body language expressions and interactions with the virtual environment, among other aspects, to enable more immersive experiences for participants. We believe these enhancements will also benefit trainees by helping them learn to identify other non-verbal expressions and interactions of individuals.

As a major future research direction, based on our observation and insights from this study, we are considering further incorporating the role of job coaches into the communication skills training program facilitated by the application. This aims to develop tools for job coaches to intervene in the training experience, in addition to their role in observing and instructing trainees during the

training process. Here, we aim to enable job coaches to guide the chatbot's behaviors and eliminate unrealistic responses in subsequent interactions by sending assistant prompts to the GPT models. This proposed feature could also be beneficial in other scenarios of communication skills training, such as interview training.

7 CONCLUSION

In this work, we developed a speech-based VR application, which leverages virtual characters as chatbots powered by LLMs, specifically using the GPT-3.5 Turbo model. With this tool, job coaches can easily create scenarios to train job-related communication skills for autistic individuals. To evaluate its user experience, we conducted a preliminary study involving three autistic trainees and two job coaches. In this study, the trainee participants interacted with the VR chatbot application in two scenarios: taking orders in a coffee shop and handling item return complaints in a market. Concurrently, the job coach participants observed and guided the trainees' experience. Based on observation and participants' feedback, our results suggest the potential and challenges of the VR chatbot application in job communication skills training. Furthermore, we identified the current utility and areas for future enhancement to optimize its job training experience.

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